

FULL TECHNICAL REPORT — AAST Unified Scraping Dataset (Final Version)

Introduction

During the development of the AAST AI System (Knowledge Graph + RAG), a full scraping pipeline was executed on the official AASTMT website. The final unified dataset contains 737 structured entries combining HTML, API data, static pages, dynamic Playwright captures, attachments, seeds, and metadata.

1. aast_v2_output/ — Full Folder Analysis

This folder contains all raw scraping outputs including files, HTML snapshots, metadata, results, and structured dumps.

1.1 files/

Contains attachments such as PDFs, images, and documents. These become File nodes in the Knowledge Graph connected via HAS_ATTACHMENT relations.

1.2 html/

Contains fully-rendered Playwright HTML, preserving dynamic menus, hidden sections, AJAX-loaded content, course lists, roadmaps, and grading systems.

1.3 json/meta.json

Metadata for each crawled page: URL, final URL, status code, depth, referrer, and timestamp. Used to reconstruct link graph edges.

1.4 json/results.json (MOST IMPORTANT)

Main structured dump: titles, text, page IDs, program IDs, staff IDs, semantic sections, images, PDF links. Used to build KG nodes & relations.

2. api responses/

Contains backend API outputs such as program data, CV data, page content, and hidden AJAX endpoints. Most structured and reliable content source.

3. playwright captures/

Contains dynamic HTML + JSON states after interactions (hover, AJAX loading). Critical for extracting hidden navigation elements.

4. static pages/

Fallback static pages including bylaws, policy pages, news, and contact info.

5. aast_records.jsonl

Preprocessed unified records ideal for semantic indexing and KG ingestion.

6. api_index.json

Maps page_id, program_id, staff ser → API JSON. Vital for linking URLs to clean backend data.

7. discover_seed_links.json

Contains 5,646 discovered URLs (programs, staff, departments, PDFs, static/dynamic pages). This is the ground truth of AAST web content.

8. static_index.json

Lists all static pages detectable without JS execution.

Unified Dataset Overview

The unified dataset contains 737 clean records: 446 HTML pages + 291 API pages.

Dataset Schema

Each record includes: source, type (HTML/API), title, and cleaned text. Ideal for RAG, semantic search, entity extraction, and KG population.

Insights From the Dataset

HTML pages contain mission, strategy, campuses, colleges, events, student life, rankings, policies, training, and announcements. API pages contain program/curriculum data, course lists, staff CVs, metadata, and contact info.

Why This Dataset Is Perfect for the AI System

It enables building a complete university KG and powering RAG applications such as student assistants, staff lookup, course recommendations, program navigation, and voice-enabled university search.

